**Specified Concept: Change**

**Related Concepts: Transformation, Energy**

https://www.youtube.com/watch?v=x49BtB5d0wg

**Physical change** is a change in the **appearance** or **form** of a substance, **without changing what it is made of**.

- **No new substance** is formed.
- It is often **reversible**.
- The **particles** of the substance stay the same, but their **arrangement** or **state** might change.

**Examples of Physical Changes:**

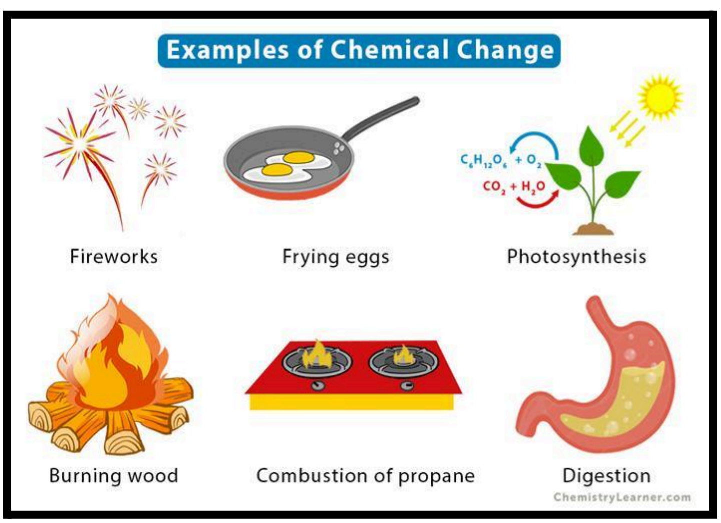
| Change | Description |
|---|---|
| Melting Ice | Ice (solid) turns into water (liquid), but it's still $H_2O$ |
| Boiling Water | Water changes to steam (gas), but it's still water |
| Breaking Glass | The shape changes, but it's still glass |
| Dissolving Sugar | Sugar disappears in water, but no new substance is formed |

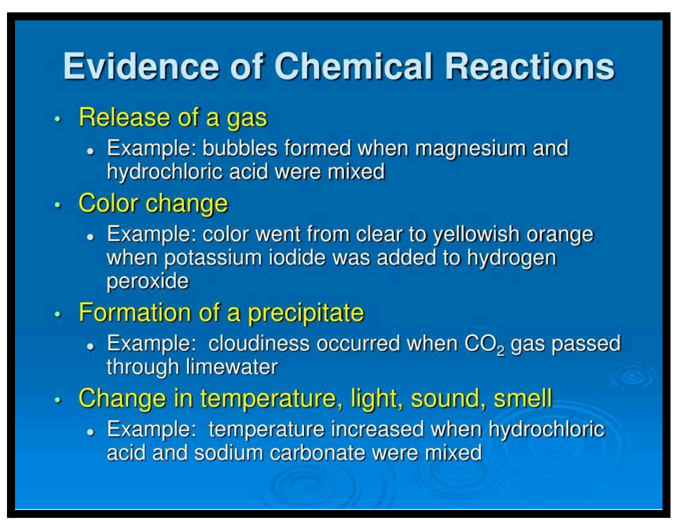
**chemical change** is a change where a **new substance is formed** with different **properties**.

- **Cannot be easily reversed**.
- There is often a **change in color, temperature, smell**, or **gas formation**.
- The **particles rearrange** to make something **new**.

**Examples of Chemical Changes:**

| Change | Description |
|---|---|
| Rusting Iron | Iron reacts with oxygen to form rust (a new substance) |
| Burning Wood | Wood turns to ash, smoke, and gas |
| Baking a Cake | Ingredients mix and change with heat to make something new |
| Vinegar + Baking Soda | Bubbles and fizzing indicate gas is formed (new substance) |



**How to Tell the Difference:**

| Feature | Physical Change | Chemical Change |
|---|---|---|
| New substance? | No | Yes |
| Reversible? | Often | Usually not |
| Examples | Melting, cutting, boiling | Burning, rusting, cooking |
| Particle Change | Rearranged, not changed | New particles formed |

## Worksheet: Physical and Chemical Changes

| Column-A | Column-B |
|---|---|
| Physical change | A change where new substance is formed |
| Chemical change | Turns iron brownish orange after exposure to air |
| Rusting | A change in state, shape or size only |
| Irreversible | A change that cannot be undone easily |
| Reversible | A change that can be done |

### Part B: Classify the Changes

Write **P** for Physical Change or **C** for Chemical Change.

1. ___ Melting butter
2. ___ Burning a candle
3. ___ Cutting paper
4. ___ Cooking an egg
5. ___ Freezing juice
6. ___ Mixing vinegar and baking soda

**Real-Life Thinking: Look** around your home or school. Write one real-life example each of:

1. A physical change you saw today: ___________________________
2. A chemical change during cooking: ___________________________
3. A change that could be both physical and chemical (Explain):

**Why is it important to know the difference between physical and chemical changes in real life?**